

Substitution

Example

If $a = 4$, $b = 2$ and $c = 3$, what are the values of the following expressions?

- a. $a + b$
- b. $b - c$
- c. $ab + c$

When we substitute, we replace something or someone with another thing or person. For example, during a football match, a manager may substitute a player by swapping a player on the field with a player on the subs' bench.

For our substitution here, we need to substitute a **letter** for a **number**.

Step 1

When we substitute, we replace the letter for its value.

We are told that, in this example, a has a value of 4. Therefore, we can substitute a for 4.

- a. $4 + b$
- b. $b - c$
- c. $4b + c$

Step 2

Now substitute the letter b for its value.

b has a value of 2:

- a. $4 + 2$
- b. $2 - c$
- c. $4b$ is slightly different; we cannot simply place the 2 next to the 4, i.e. the answer is not $42 + c$.

Rather, we must remember that invisible sign between the 4 and b . So, our substitution becomes 4×2 which is equal to 8.

$$8 + c$$

Step 3

Substitute the letter c for its value.

- a. $4 + 2$
- b. $2 - 3$
- c. $8 + 3$

Step 4

You have substituted the values for the letters you have been given.

- a. $4 + 2 = 6$
- b. $2 - 3 = -1$
- c. $8 + 3 = 11$